

Clustering: Exercises

Exercise

Consider a pointset P of N points represented as a key-value pairs (ID_x, x) , where ID_x is a distinct integer in $[0, N)$, and x is a point. Let also S be a set of k centers. Assuming $k \leq \sqrt{N}$ and considering S as global data, show that $\text{Assign}(P, S)$ can be implemented in MapReduce in 1 round using $M_L = O(\sqrt{N})$ and $M_A = O(N)$.

Input $\{(ID_x, x) : ID_x \in [0, N) \text{ distinct}, x \in P\}$
 $S = \{c_1, c_2, \dots, c_k\}$ GLOBAL VARIABLE

Output $\{(ID_x, (x, c(x))) : ID_x \in [0, N) \text{ distinct}, x \in P$
 $c(x) = \text{center of } S \text{ closest to } x\}$

SIMPLEST SOLUTION

Round 1

Map Phase $(ID_x, x) \rightarrow (ID_x, (x, c(x)))$

$c(x)$ computed as $\arg\min_{c \in S} d(x, c)$

We are exploiting the fact that S is a global variable

Reduce Phase: empty

$M_L = O(k)$ assuming that each point and each center takes constant space

$M_A = O(N)$ $N = |P|$

if we assume that the copies of S created are not too many, say $O(N/k)$

More conservative implementation of Assign(P, S)

Round 1

Map Phase $(ID_x, x) \rightarrow (ID_x \bmod \sqrt{N}, (ID_x, x))$

Reduce Phase For every $j \in [0, \sqrt{N})$ separately
gather all intermediate pairs with key j into
 (j, L_j) $L_j \equiv$ List of values (ID_x, x) from
these pairs $|L_j| = O(\sqrt{N})$

for each (ID_x, x) in L_j return
 $(ID_x, (x, c(x)))$ $c(x) = \arg \min_{c \in S} d(x, c)$

$$M_L = O(\sqrt{N} + k) = O(\sqrt{N})$$

$M_A = O(N)$ no replication of the
input points and only $\sqrt{N}$ copies
of S are made $\Rightarrow \sqrt{N} \cdot k$ is
the aggregate space taken by these copies
 $\sqrt{N} \cdot k = O(N)$

Exercise

Let P be a set of N points in a metric space (M, d) , and let $T \subseteq P$ be a coreset of $|T| > k$ points such that for each $x \in P$ we have $d(x, T) \leq \epsilon \Phi_{k\text{center}}^{\text{opt}}(P, k)$, for some $\epsilon \in (0, 1)$. Let S be the set of k centers obtained by running the Farthest-First Traversal algorithm on T , and let C_S be the clustering returned by $\text{Assign}(P, S)$. Prove an upper bound to $\Phi_{k\text{center}}(C_S)$ as a function of ϵ and $\Phi_{k\text{center}}^{\text{opt}}(P, k)$.

Explores a generalization of the analysis done for MR-Farthest-First Traversal, where ϵ was equal to 2

To solve the exercise, adapt the proof of the Theorem in Slides 35÷38 of the Slides on Clustering (Part 2)

Exercise

Let P be a set of points in a metric space (M, d) , and let $T \subseteq P$. For any $k < |T|, |P|$, show that $\Phi_{\text{kcenter}}^{\text{opt}}(T, k) \leq 2\Phi_{\text{kcenter}}^{\text{opt}}(P, k)$. Is the bound tight?

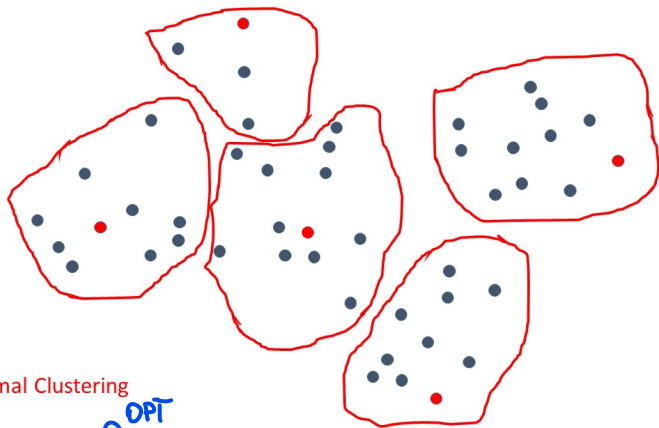
The exercise implies that the quality of the best k -center clustering of T is not much worse than the quality of the best k -center clustering of P .

How to solve the exercise?

IDEA : find a "good" k -clustering of T as a PROJECTION of the optimal k -clustering of P .

$\mathcal{C}_P^{\text{OPT}} = (C_1, C_2, \dots, C_k; c_1, c_2, \dots, c_k)$ OPTIMAL
k-center clustering of P

- * Remove from each C_i the points not in T (i.e., project $\mathcal{C}_P^{\text{OPT}}$ on T)
- * Select one arbitrary point from each $C_i \cap T$ (if any)
- * Let $S' \subseteq T$ be the set of selected points
 $\Rightarrow |S'| \leq k$



Optimal Clustering

C_p^{OPT}



Optimal Clustering after
removal of points not in T

$S' \equiv$ points in $\square$

How far is each $t \in T$ from the closest center c in S'

C_i (one of the optimal clusters)



$$d(t, c) \leq d(t, c_i) + d(c_i, c) \leq 2\Phi_{k\text{center}}^{\text{OPT}}(P, k)$$

$$\Rightarrow d(t, S') \leq 2\Phi_{k\text{center}}^{\text{OPT}}(P, k)$$

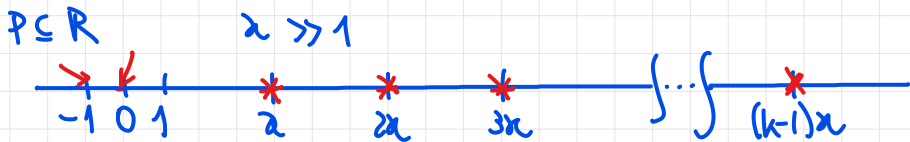
ALTERNATIVE ARGUMENT: Run Farthest-First Traversal on T to select a set S' of k centers, and argue that for each $t \in T$ $d(t, S') \leq 2\Phi_{k\text{center}}^{\text{OPT}}(P, k)$, reasoning as in slide 36 of the slides on Clustering (Part 2)

Let $P_{T,S'}$ = clustering around S' ($\equiv \text{Assign}(P, S')$)

we have

$$\Phi_{\text{kenter}}^{\text{opt}}(T, k) \leq \Phi_{\text{kenter}}(P_{T,S'}) \leq 2 \Phi_{\text{kenter}}^{\text{opt}}(P, k)$$

Is the bound tight?



$$|P| = k+2$$

$$T = P - \{0\}$$

$$|T| = k+1$$

$$\Phi_{\text{kenter}}^{\text{opt}}(P, k) = 1$$

$$\Phi_{\text{kenter}}^{\text{opt}}(T, k) = 2$$

Exercise

Let P be a set of N bicolored points from a metric space, partitioned into k clusters $C_1, C_2, \dots, C_k$. Each point $x \in P$ is initially represented by the key-value pair $(ID_x, (x, i_x, \gamma_x))$, where ID_x is a distinct key in $[0, N-1]$, i_x is the index of the cluster which x belongs to, and $\gamma_x \in \{0, 1\}$ is the color of x .

- 1 Design a 2-round MapReduce algorithm that for each cluster C_i checks whether all points of C_i have the same color. The output of the algorithm must be the k pairs (i, b_i) , with $1 \leq i \leq k$, where $b_i = -1$ if C_i contains points of different colors, otherwise b_i is the color common to all points of C_i .
- 2 Analyze the local and aggregate space required by your algorithm. Your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.



Straightforward approach: gather all points of the same cluster. Problematic because it requires too much local space (in the worst case)

Better idea

- * Subdivide P into $\sqrt{N}$ partitions
- * Within each partition determine the unique color of each cluster (if any)
- * Merge the information for each cluster ($\leq \sqrt{N}$ pieces to merge for each cluster)

Round 1

Map Phase $(ID_x(x, i_x, r_x)) \rightarrow (ID_z \bmod \sqrt{N}, (x, i_x, r_x))$

Reduce Phase for each key $j \in [0, \sqrt{N})$ separately
gather all intermediate pairs with key j into
 (j, L_j) where $L_j \equiv$ list of values (x, i_x, r_x) from
these pairs. For each cluster index i s.t.
 $\exists (x, i_x = i, r_x)$ in L_j return a pair
 $(i, b_i(j))$

$b_i(j) = -1$ if L_j contains points from cluster i with different colors

$b_i(j) = \text{common color of points in } L_j \text{ from cluster } i$ (if they all have the same color)

$\Rightarrow$ At the end of Round 1 there will be $\leq \sqrt{N}$ pairs $(i, b_i(j))$ for each i

Round 2

Map Phase: empty

Reduce Phase For each $1 \leq i \leq k$ separately
 gather all pairs $(i, b_i(j))$ $j \in [0, \sqrt{N})$
 into (i, L_i) where $L_i \equiv$ list of values $b_i(j)$ from
 these pairs. Return (i, b_i) where

$$b_i = b \quad \text{if } b_i(j) = b \quad b_i(j) \in L_i$$

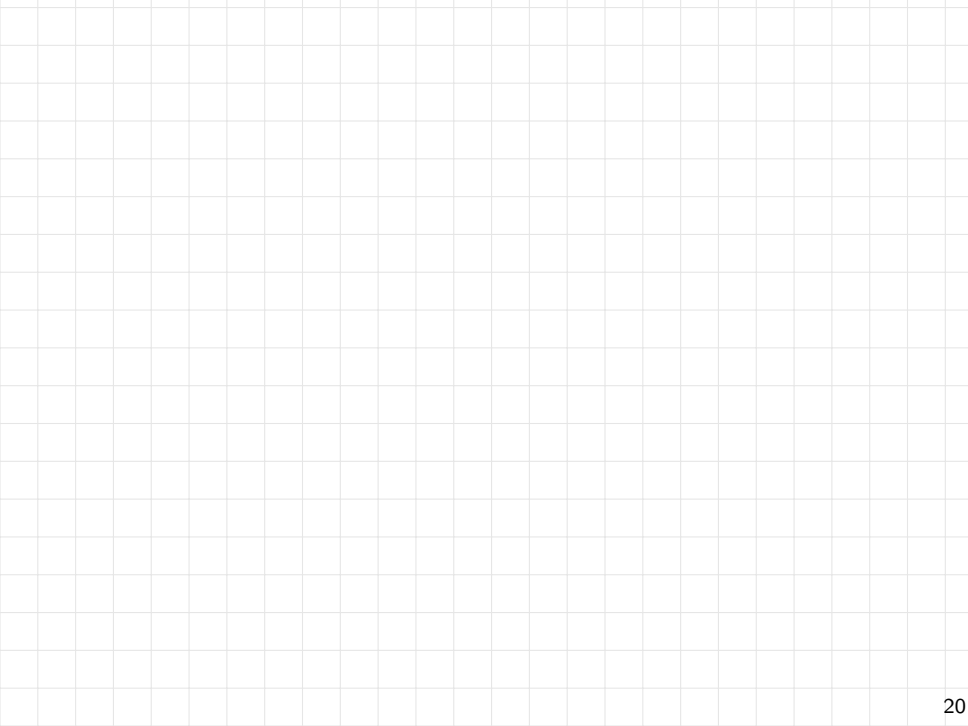
$$b_i = -1 \quad \text{otherwise}$$

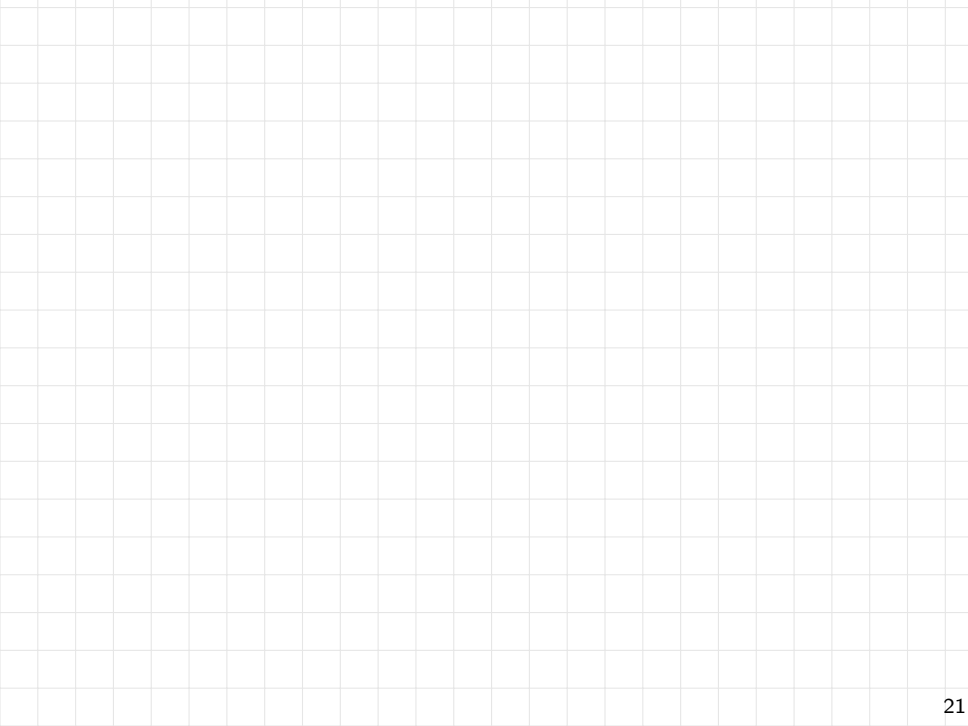
$$M_L = O(\sqrt{N})$$

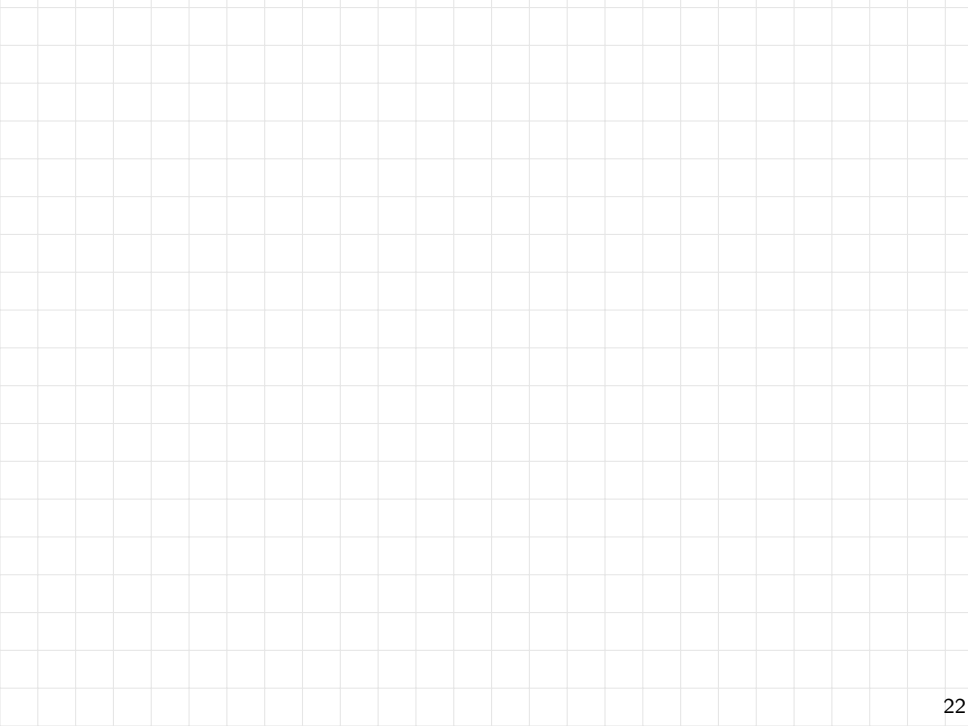
$$M_A = O(N)$$

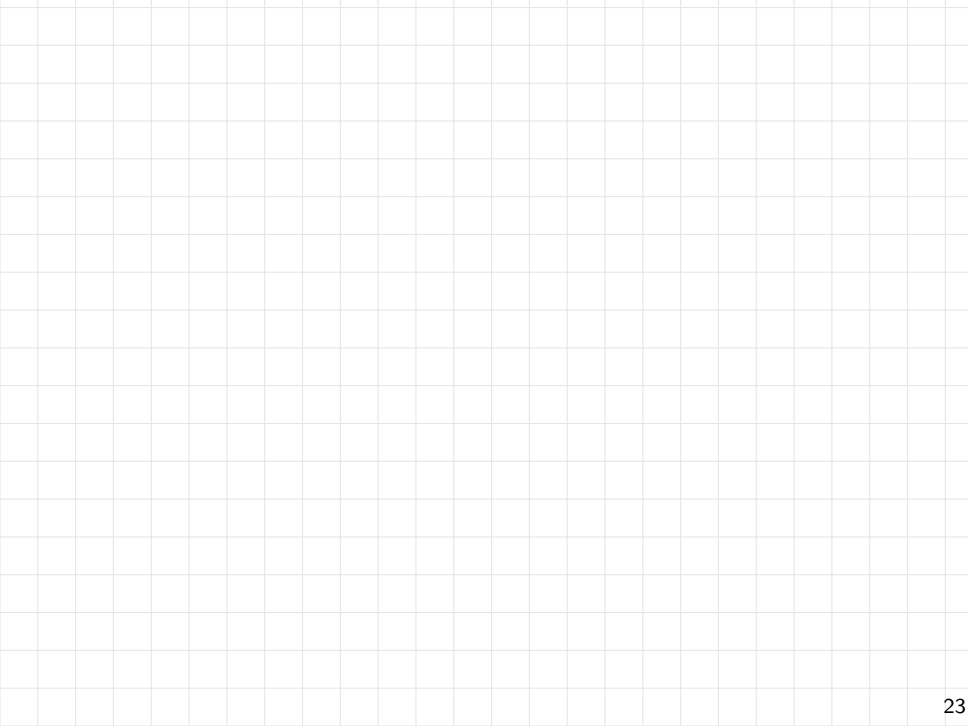
Exercise

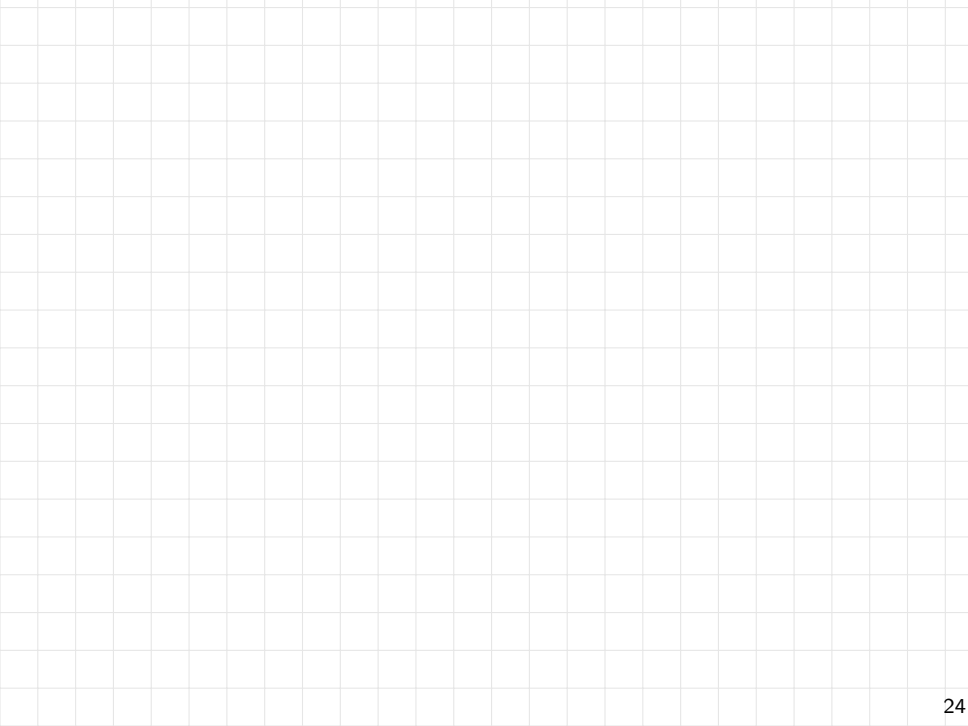
Show how to implement Lloyd's algorithm in MapReduce using $O(1)$ rounds per iteration, local space $M_L = O(\sqrt{N})$ and aggregate space $M_A = O(N)$. (**Hint:** Make the set of centers and Φ global variables.)

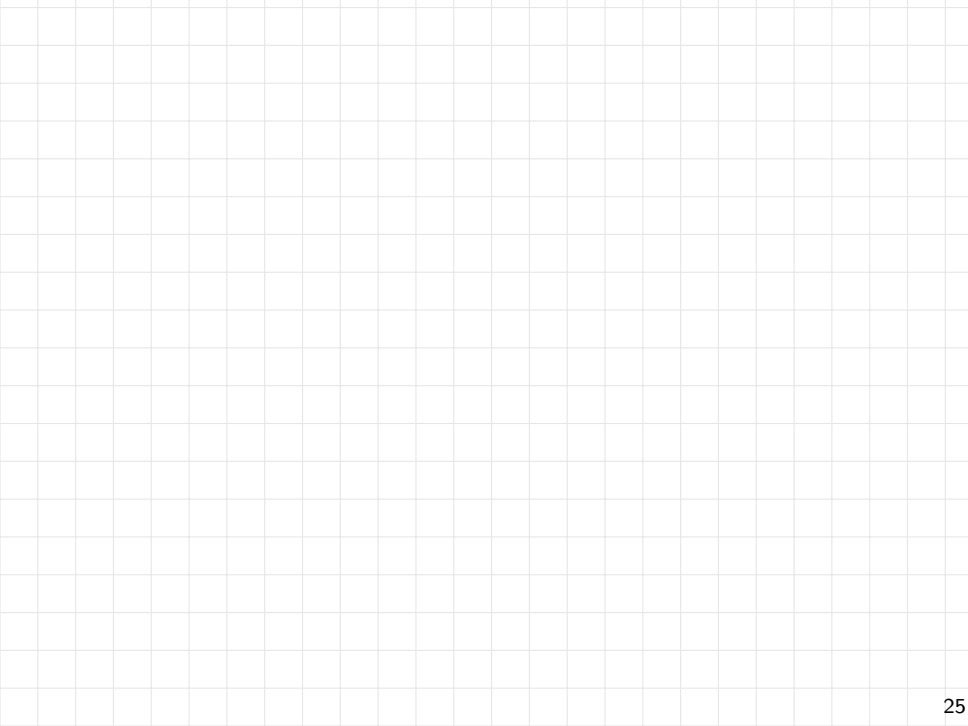


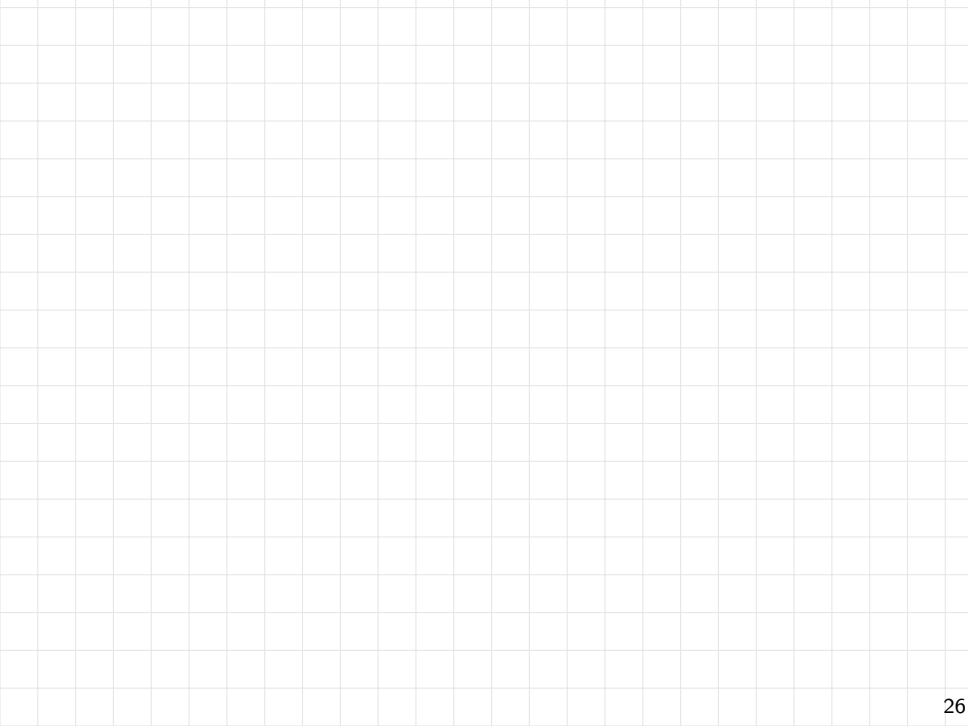












Exercise

Let $\mathcal{C}_{\text{alg}}$ be the k -clustering of P induced by the set S of centers returned by k-means++ (i.e., $\mathcal{C}_{\text{alg}} = \text{Assign}(P, S)$). Suppose that we know that for some value $\alpha > 1$ the following probability bound holds:

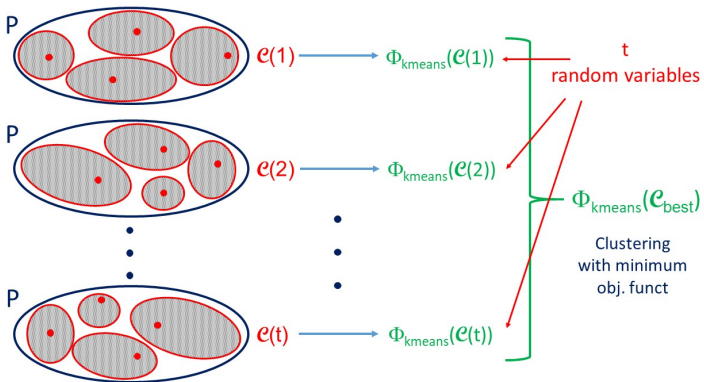
$$N = |P|$$

$$\Pr(\Phi_{\text{kmeans}}(\mathcal{C}_{\text{alg}}) \leq \alpha \cdot \Phi_{\text{kmeans}}^{\text{opt}}(P, k)) \geq 1/2.$$

Show that the same approximation ratio, but with probability at least $1 - 1/N$, can be ensured by running several *independent instances* of k-means++, computing for each instance the clustering around the returned centers, and returning at the end the best clustering found (call it $\mathcal{C}_{\text{best}}$), i.e., the one with smallest value of the objective function.

Consider the following setting

- Run t independent instances of k-means++ on input P .
- $\mathcal{C}(i)$ = clustering around the centers returned by i -th instance.
- $\mathcal{C}_{\text{best}}$ = best clustering among $\mathcal{C}(i)$'s. (i.e., smallest obj. function)



Let's call $\text{OPT} = \Phi_{\text{kmeans}}^{\text{OPT}}(P, k)$

We must show that $\text{Prob}(\Phi_{\text{kmeans}}(\mathcal{C}_{\text{best}}) \leq \alpha \text{OPT}) \geq 1 - \frac{1}{N}$
when t is suitably large

Define the event $E_i = \Phi_{\text{kmeans}}(\mathcal{C}(i)) > \alpha \text{OPT}$

By hypothesis: $\text{Prob}(E_i) \leq 1/2 \quad \forall i$

$$\begin{aligned}
 \text{Now } \Pr(\Phi_{\text{kmeans}}(\rho_{\text{best}}) \leq \alpha \text{OPT}) &= \\
 1 - \Pr(\Phi_{\text{kmeans}}(\rho_{\text{best}}) > \alpha \text{OPT}) &= \\
 1 - \Pr(E_1 \cap E_2 \cap \dots \cap E_t) &= \\
 1 - (\Pr(E_1))^t &\geq 1 - (1/2)^t
 \end{aligned}$$

If we fix $t = \lceil \log_2 N \rceil$

$$\Rightarrow \Pr(\Phi_{\text{kmeans}}(\rho_{\text{best}}) \leq \alpha \text{OPT}) \geq 1 - \frac{1}{N}$$